

Now, intuitively, if we can devise our security primitives and constructs in that zone (i.e., the green coloured zone in Figure 4), then we are safe. Well, we are right, but only partially. Having security primitives in the green zone in Figure 4 is a necessary but not a sufficient condition to prove the quantum safety of the solution.

There are believed to be some security techniques available today that are promising in post quantum order, e.g., secret-key cryptography. However, the quantum property of the quantum computer, raises an additional security attack surface as well. One such example is the superposition attack. To put it in simple terms, superposition is a quantum property where a qubit (a bit in a quantum computer is called qubit) can pose 0, 1 or a mix of 0 and 1 values. Once you measure the qubit, it collapses to either classical 0 or classical 1. Now an adversary, powered by the quantum computer oracle, can potentially learn the superposition ciphertexts of some plain text and decipher the superposition using another algorithm (without directly measuring the superposition ciphertext) to acquire knowledge about the cryptosystem. That's it.

One can, of course, argue that to pose such a threat to our contemporary, classical security primitives, one needs to have a big enough (of many qubits), fault-tolerant and stable quantum computer. This will happen in the near future, but is not happening tomorrow. Then why should I have to bother about it, right now? The primary reasons are: an adversary can potentially steal your encrypted messages of today and decipher them tomorrow, powered by the quantum oracle. So this sort of development requires a holistic research approach and an overhaul of the cryptographic infrastructure, which are time consuming. So it's better to start today.

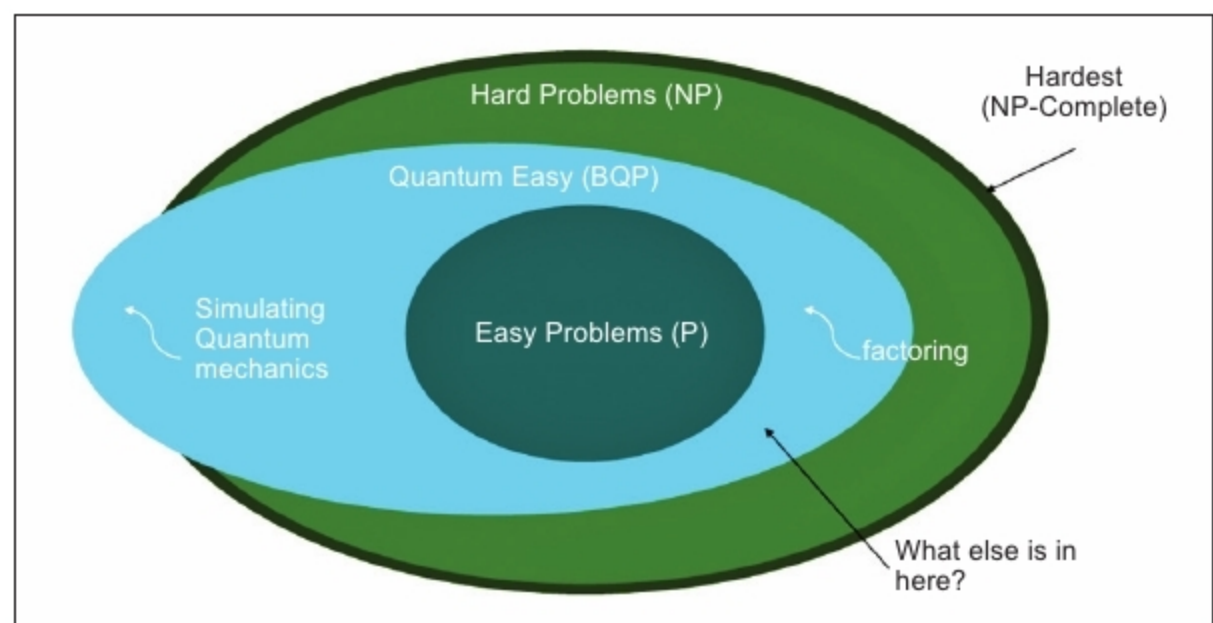


Figure 4: BQP solvability of quantum computers

Quantum gadgets

Is the picture painted very gloomy? Not really. There is a silver lining. We now have something called a quantum gadget which can be used to enhance the security of the classical communication primitives. One such example is quantum key distribution (QKD). The idea of QKD is derived from the fact that two honest parties can have a shared random secret key only known to them. If any adversary wants to intercept the secret random key, the adversary must read the key. Once a quantum state is read (i.e., measured) it collapses to one of the classical states – either 0 or 1. Hence any such adversary intrusion results in a detectable anomaly in the overall system.

There are other ways in which a quantum gadget can pave the way into our classical security systems and primitives, e.g., quantum fingerprints. One can think

of it as a technique involving a quantum computer to generate a string like our classical cryptographic hash functions. There are various other examples also, but not limited to this—quantum random number generation, quantum signature generation, Byzantine agreement, quantum flip-the-coin, secure e-voting, secure multi-party computation, *et al.*

In a nutshell

Quantum computer research, as shown in Figure 2, is widespread across nations. One of the areas of immediate interest for research is related to making classical devices, communication and information to be adaptable and secure in the post quantum era. Basically, it means making them quantum safe. Of course, the journey has just started, so we are far from reaching any definitive answers. **END** 🐧

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